

Mediation Analysis

STSCI / INFO / ILRST 3900: Causal Inference

Sep 31 2026

Learning goals for today

At the end of class, you will be able to:

1. Describe how mediation analysis can help make better decisions
2. Understand how the total effect decomposes into a direct and indirect effect

Logistics

- ▶ Quiz 2: Oct 8

How do treatments work?

- ▶ So far:
Does assigning individuals a treatment cause a change in the outcome?
- ▶ Measured through ATE

$$E(Y^{a=1}) - E(Y^{a=0})$$

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- ▶ Today:
How/why does assigning a treatment cause a change in the outcome?
- ▶ Mediation analysis

Example: Effect of ChatGPT on math test performance¹

A : ChatGPT Use $\longrightarrow$ Y : Math Score

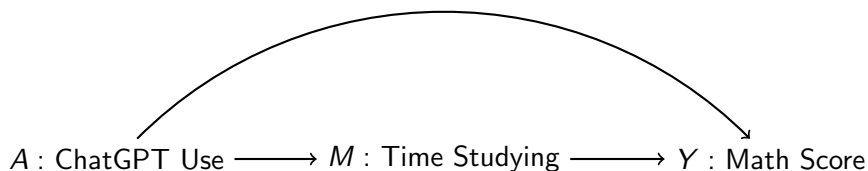
¹Bastani et al (2025). "Generative AI without guardrails can harm learning: Evidence from high school mathematics. " Proceedings of the National Academy of Sciences

Example: Effect of ChatGPT on math test performance¹

$A : \text{ChatGPT Use} \longrightarrow M : \underline{??} \longrightarrow Y : \text{Math Score}$

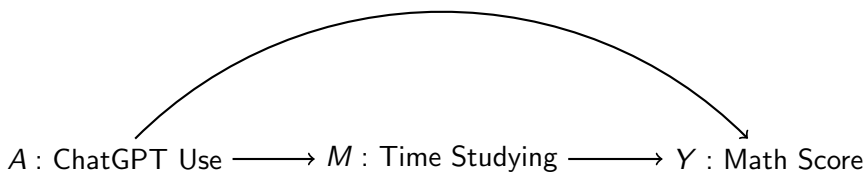
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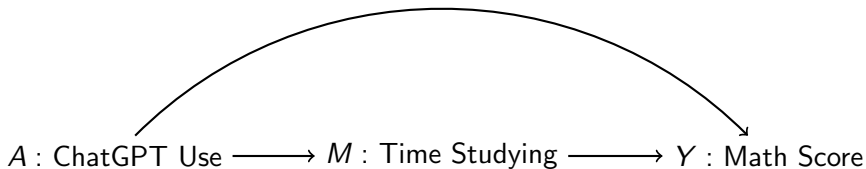
- What is the causal effect of ChatGPT on Math scores?

Example: Effect of ChatGPT on math test performance



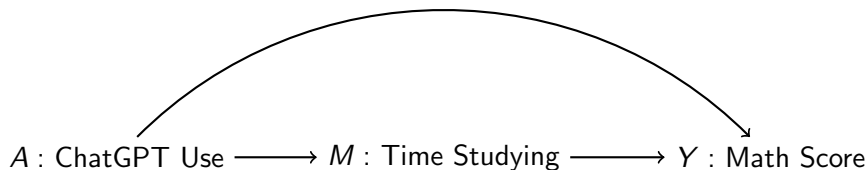
- ▶ What is the causal effect of ChatGPT on Math scores?
- ▶ What is the causal effect of ChatGPT on Math scores due to changing studying time?
- ▶ What is the causal effect of ChatGPT on Math scores due to other things?

Example: Effect of ChatGPT on math test performance



- ▶ What is the causal effect of ChatGPT on Math scores?
- ▶ What is the causal effect of ChatGPT on Math scores due to changing studying time?
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- ▶ Time studying is a **mediator** (i.e., middleman, go-between)

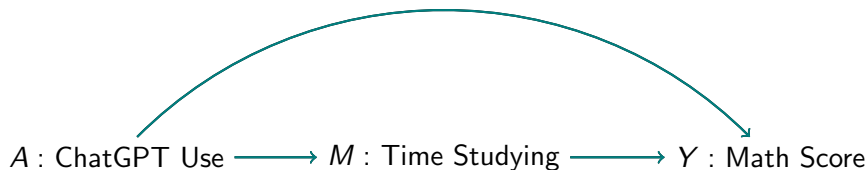
Types of effects



- **Total effect** of ChatGPT

$$ATE = E(Y^{a=\text{chatGPT}}) - E(Y^{a=\text{No ChatGPT}})$$

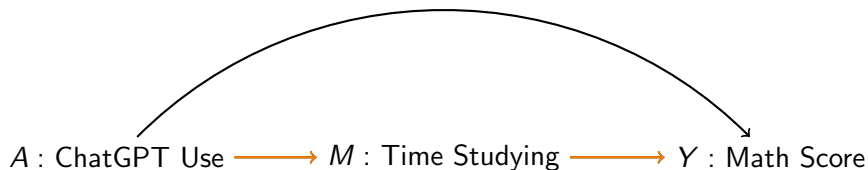
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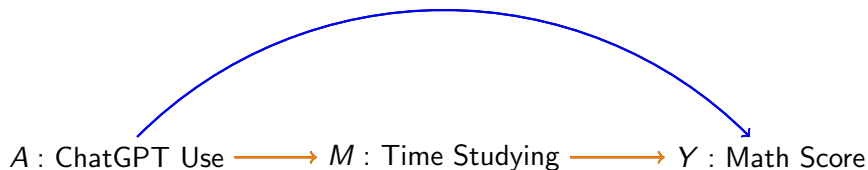


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- **Indirect effect**: effect due to studying time

Types of effects

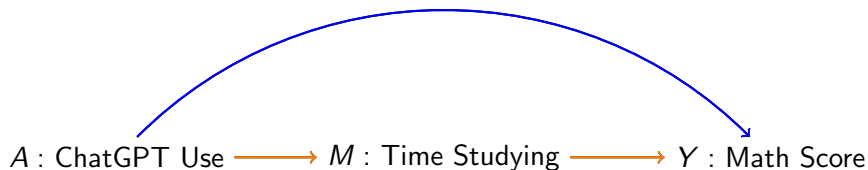


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- ▶ **Indirect effect**: effect due to studying time
- ▶ **Direct effect**: effect due to everything else
- ▶ Direct effect is defined with respect to a mediator (i.e., cause from everything else besides the specified mediator)

How would this shape decisions?

Observe: Total effect of ChatGPT on math scores is negative

$$ATE = E(Y^{a=1}) - E(Y^{a=0}) < 0$$

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What if:

- ▶ Indirect effect: ↓
ChatGPT decreases study time and which subsequently decreases math scores
- ▶ Direct effect: ↑
ChatGPT increases ability to guess well which subsequently increases math scores
- ▶ Total effect: ↓
First effect outweighs second effect so total effect is negative

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If you wanted to do well on the test, you could use ChatGPT to increase your ability to guess well, but still force yourself to study the same amount

Mediation analysis

Why is mediation analysis useful?

- ▶ Understanding why treatments work is of intrinsic value
- ▶ Guides how to modify or change treatment:

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- ▶ Understanding why treatments work is of intrinsic value
- ▶ Guides how to modify or change treatment:
 - ▶ Can we modify treatment to make effect stronger?
 - ▶ Can we modify treatment to mitigate side effects of treatment?
 - ▶ Can we modify treatment to get same effect but at lower cost?

Notation

- Potential outcome for treatment:

$$Y_i^{a=1}$$

Outcome we would observe if individual i is assigned treatment value 1

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- Potential outcome for treatment and mediator:

$$Y_i^{a=1, m=1}$$

Outcome we would have observed if individual i is assigned treatment value 1 **and is assigned mediator value 1**

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Outcome we would have observed if individual i is assigned treatment value 1 **and is assigned mediator value 1**

- Potential mediator for treatment:

$$M_i^{a=1}$$

The mediator we would have observed if individual i is assigned treatment value 1

Notation practice

► $Y_i^{a=\text{chatGPT}, m=3}$

► $Y_i^{a=\text{chatGPT}, m=5}$

► $Y_i^{a=\text{NochatGPT}, m=5}$

► $M_i^{a=\text{chatGPT}}$

Notation practice

► $Y_i^{a=\text{chatGPT}, m=3}$

Math score we would observe if individual i is assigned to use chatGPT and assigned to study for 3 hrs

► $Y_i^{a=\text{chatGPT}, m=5}$

Math score we would observe if individual i is assigned to use chatGPT and assigned to study for 5 hrs

► $Y_i^{a=\text{NochatGPT}, m=5}$

Math score we would observe if individual i is assigned to **not** use chatGPT and assigned to study for 5 hrs

► $M_i^{a=\text{chatGPT}}$

The number of hours individual i would have studied if assigned to use ChatGPT

Controlled Direct Effect

Controlled direct effect: What is the effect of the treatment when fixing the mediator to a specific value m ?

$$CDE \text{ for level } m = E(Y^{a=1,m}) - E(Y^{a=0,m})$$

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► Example:

$$CDE \text{ for studying 3 hrs} = E(Y^{a=1,m=3}) - E(Y^{a=0,m=3})$$

What is the causal effect of using ChatGPT when students are also assigned to study for 3 hours?

► Can be different for each possible value of the mediator

Natural Direct Effect

Natural Direct effect:

What is the effect of treatment when keeping the mediator fixed at the value it would have been without treatment?

$$NDE = \underbrace{E(Y^{a=1, m=M^{a=0}})}_{\text{PO if assigned treatment but mediator is set to what it would be without treatment}} - \underbrace{E(Y^{a=0, m=M^{a=0}})}_{\text{PO if assigned \textbf{no} treatment and mediator is set to what it would be without treatment}}$$

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- ▶ Roughly the controlled direct effect for m set to what it would have been without treatment
- ▶ $M_i^{a=0}$ can vary across individuals

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What is the effect of ChatGPT when assigning people to study the same amount they would've studied without ChatGPT?

Natural Indirect Effect

Natural Indirect effect: What is the effect keeping treatment fixed, but changing the mediator from what it would've been under treatment to what it would've been under no treatment?

$$NIE = \underbrace{E(Y^{a=1, m=M^{a=1}})}_{\text{PO if assigned treatment and mediator is set to what it would be **with** treatment}} - \underbrace{E(Y^{a=1, m=M^{a=0}})}_{\text{PO if assigned treatment but mediator is set to what it would be **without** treatment}}$$

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What is the effect when assigning everyone to use ChatGPT, but changing the amount of study time between what it would've been with/without ChatGPT?

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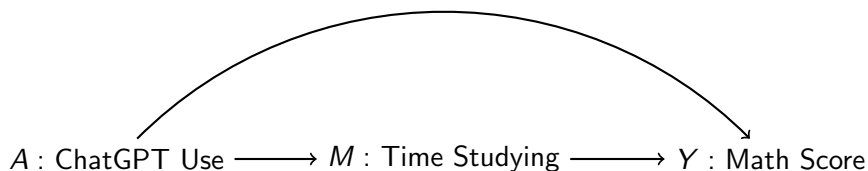
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Total Effect is the sum of natural direct and natural indirect effect

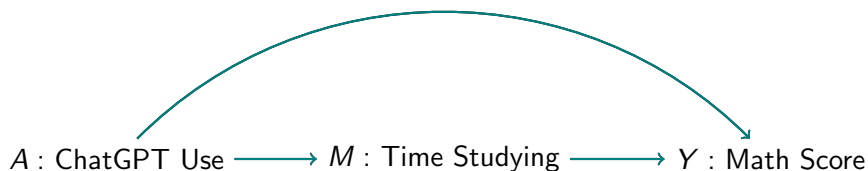
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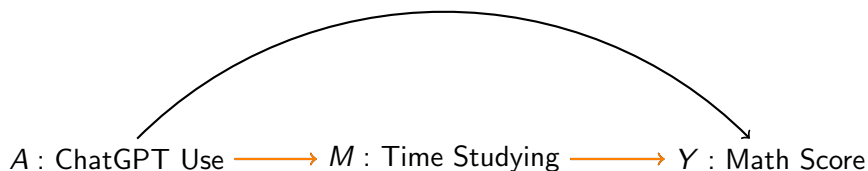
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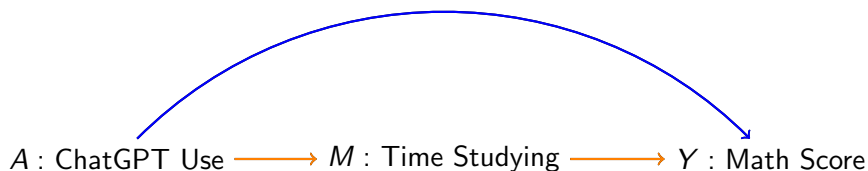


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